Opus 4.8

```
## Setting Up the Solution
The equation is linear:
$$x_3' + \tanh(t) x_3 = e^{-
  t/3}.$$
**Integrating factor:**
$$\mu(t)=\exp\left(\int\tanh
  t\,dt\right)=\exp(\ln\cosh
  t)=\cosh t.$$
```

Kimi-K2.6 (unprefilled)

```
The given first-order linear
  ODE is
$$x_3'(t)+\tanh(t)\,x_3(t)=e
  ^{-t/3}, \qquad x_3(0)=5.
  $$

**1. Solve the ODE.**
The integrating factor is
$$\mu(t)=\exp\!\left(\int
  \tanh t\,dt\right)=\exp\!
  \bigl(\ln\cosh t\bigr)=
  \cosh t.$$
```

GPT-OSS-120B (unprefilled)

```
The linear equation

\[
x_3'(t)+\tanh t\,x_3(t)=e^{-
  t/3},\qquad x_3(0)=5
\]

has integrating factor

\[
\mu(t)=\exp\!\Bigl(\int
  \tanh t\,dt\Bigr)=\exp(\ln
  \cosh t)=\cosh t .
\]
```